

Answer Key - Machine Learning Basics

Q1: What is machine learning?

Machine learning is a subset of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed. It uses algorithms to analyze data and make predictions based on patterns.

Q2: Explain supervised learning

Supervised learning is a machine learning approach where the algorithm learns from labeled training data. Each training example includes both input features and the correct output label. The model learns the mapping between inputs and outputs through this labeled data.

Q3: What is the difference between regression and classification?

Regression is used to predict continuous numerical values such as house prices or temperature. Classification is used to predict discrete categorical values such as whether something is a cat or dog. These are two different types of supervised learning problems.